Lecture 11 Mixed digital methods design of Analysis

Digital methods

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Pick up from last time.

Today's tasks

1. Different types of coding
2. Mixed methods designs recap
3. Mixed methods analysis design

Types of coding and reading

Types of Codes	Reading attitude	Learning objective	Output
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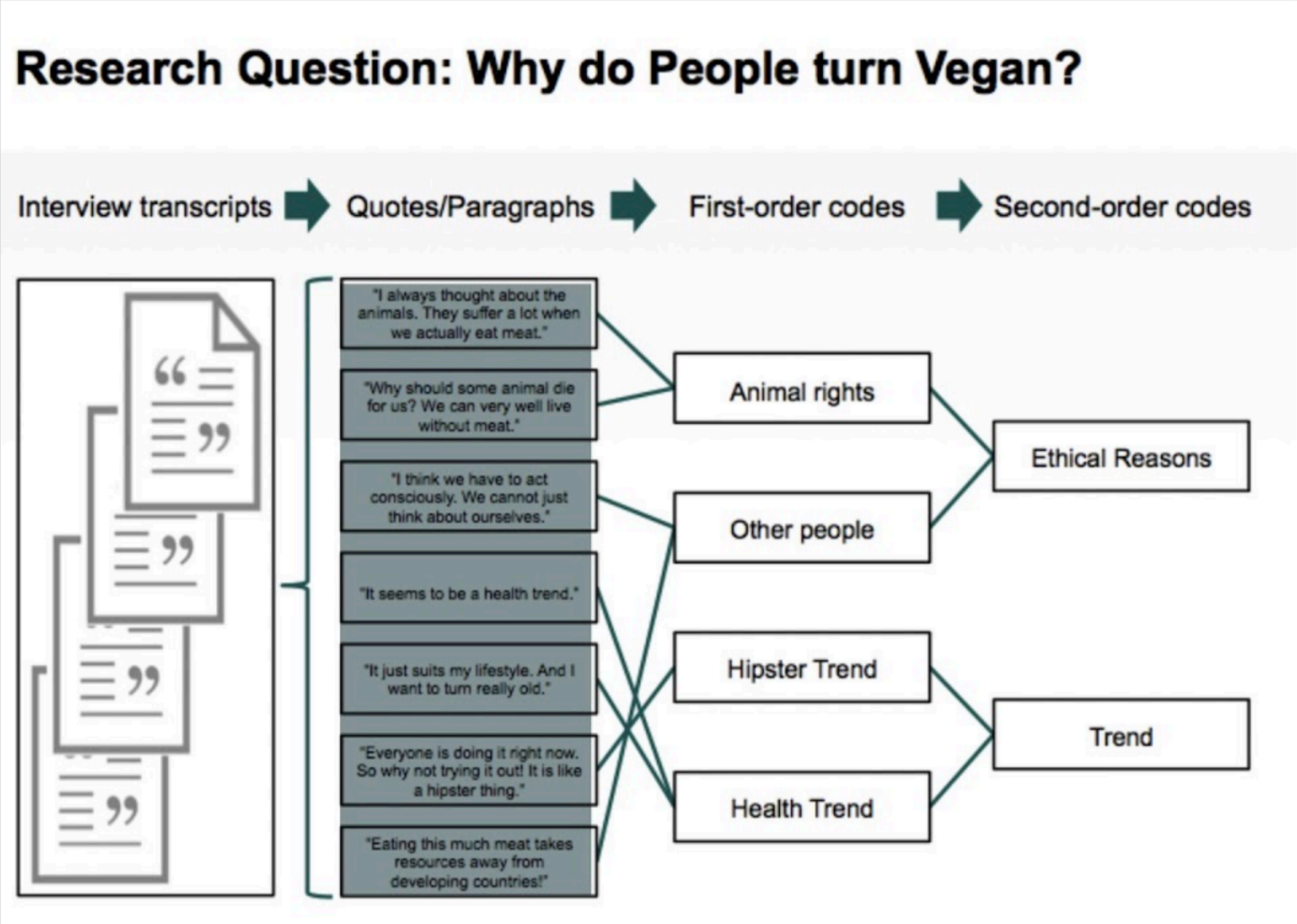
Closed coding	Valid, fair, reliable and effective	Good Quantitative Measurement	A **variable** to either estimate quantity or use as training and/ or testing of computational classification.

Types of coding and reading



Computer assistent open, focused and closed reading

Types of Codes	Reading attitude	Learning objective	Output	Type of computer assistance	Contribution of computational methods
Open coding	open, curious, explorative, fast	Extensive knowledge of themes, Locate analytic interest	Notes, memos, codes, initial coding tree	Unsupervised clustering and classification methods, such as LDA topic model or HSBM	Support or challenge initial analysis of central themes or thematic bounderies, provide initial sense of wider discursive context.
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Focused coding	Analytical, testing, theoretically invested, slow	Important internal differences, boundaries, relations and consequence	Analysis of categories, coding scheme and finale coding tree	Search queries extended with word2vec	Support document retrieval for intensive and extensive analysis of a given theme.
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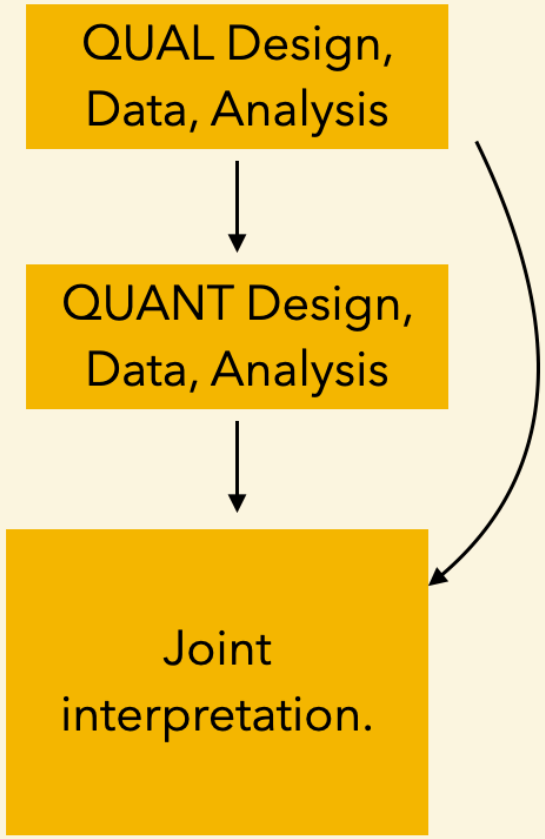
Closed coding	Valid, fair, reliable and effective	Good Quantitative Measurement	A **variable** to either estimate quantity or use as training and/ or testing of computational classification.	Dictionary classification, active/ supervised machine learning.	Enable the classification of the full dataset.

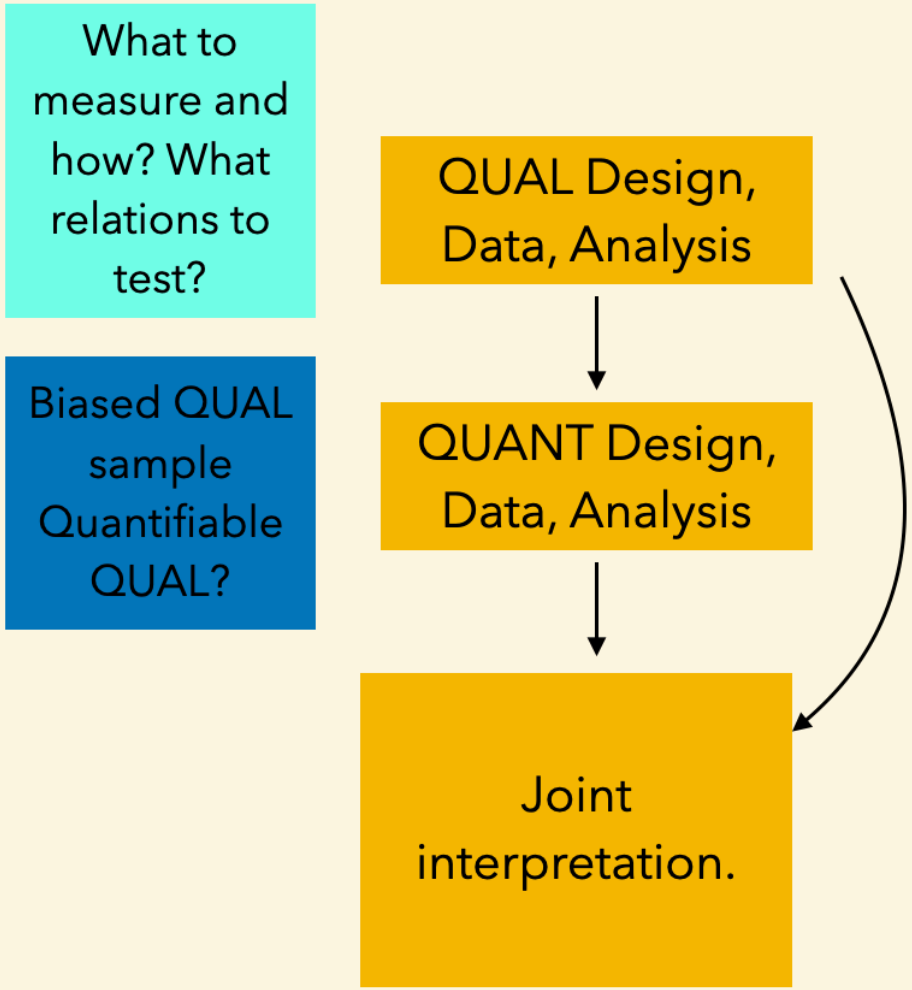
Sequencial design

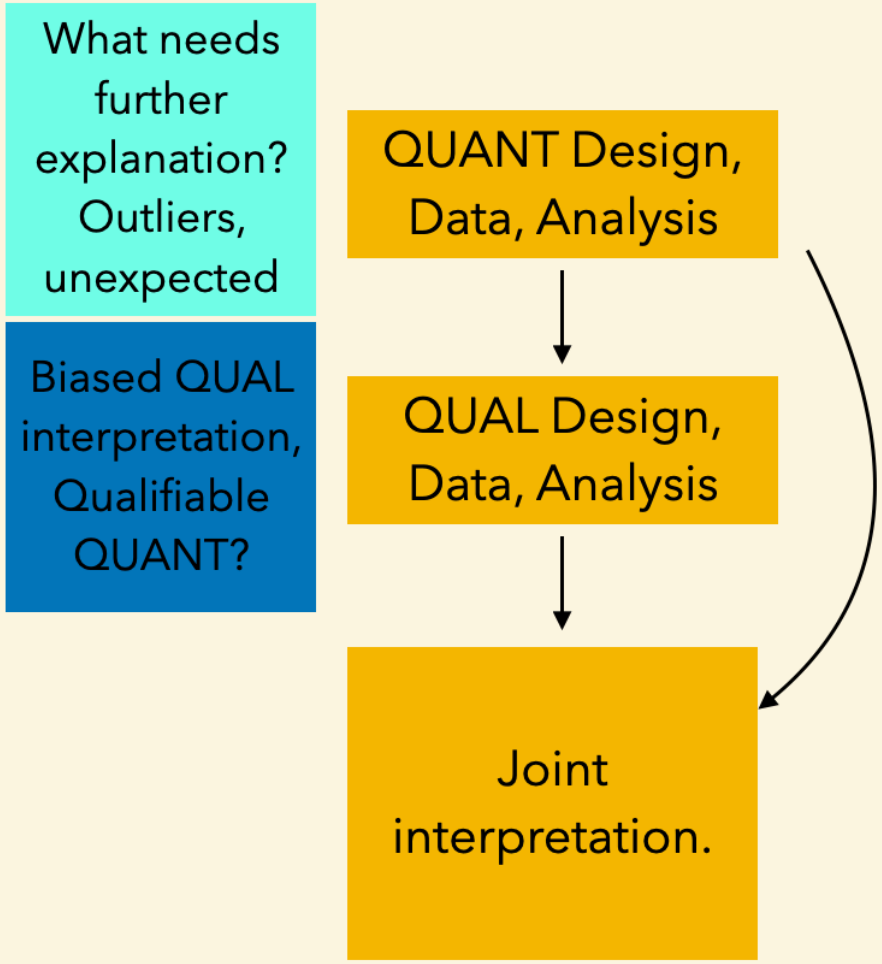
What to measure and how? What relations to test?

Biased QUAL sample Quantifiable QUAL?



Sequencial design



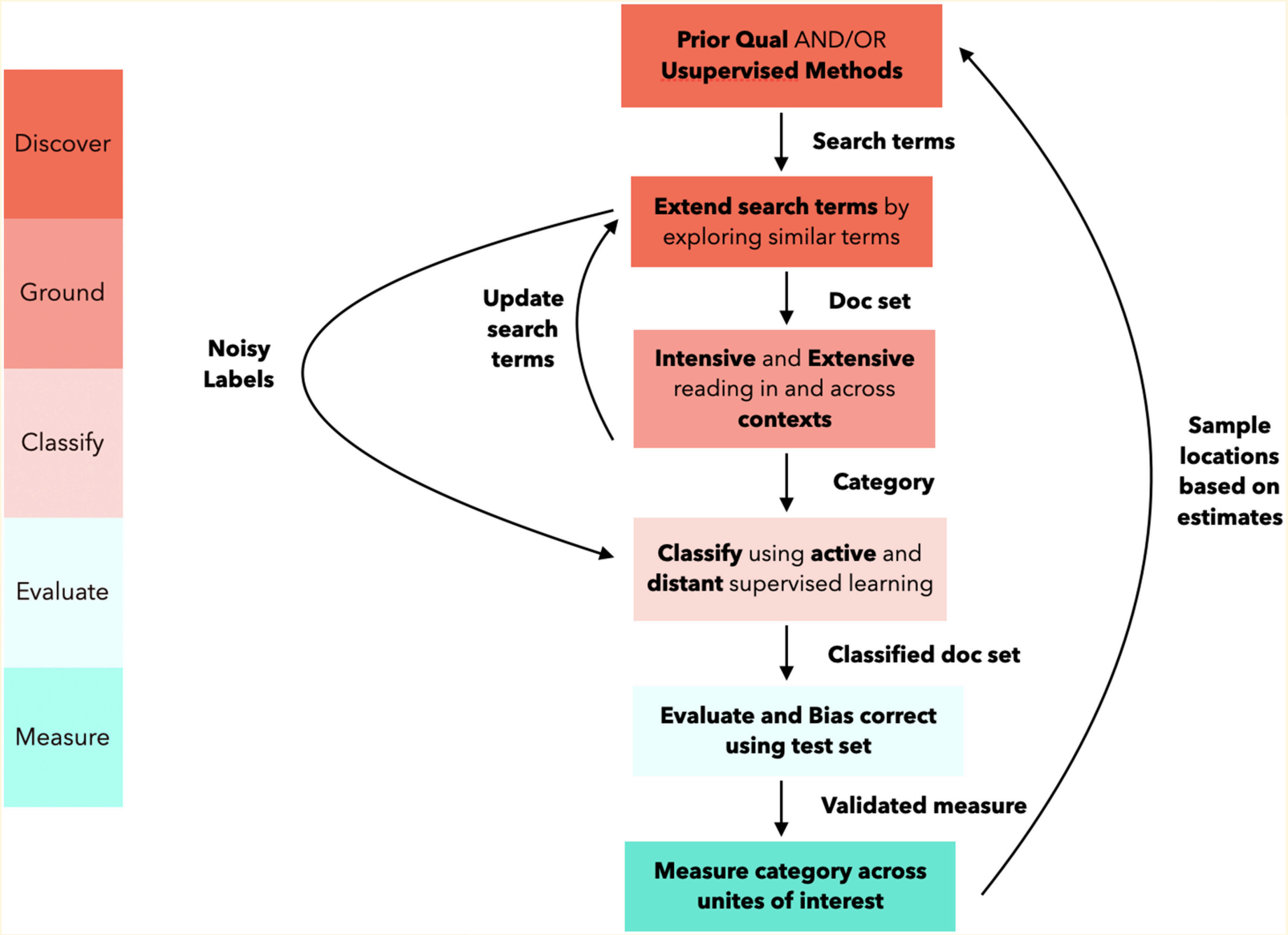


Cool thing with digital text data and computational text analysis

1. **Computational methods assist qual analysis**
    - ensures saturation
    - ensures against bias qual
    - enable disovery by alternative means

Cool thing with digital text data and computational text analysis

1. Computational methods assist qual analysis
    - ensures saturation
    - ensures against bias qual
    - enable disovery by alternative means
2. **Qual and quant analysis of same observations**
    - ensures data integration
    - ensures analytical integration
    - Allows us to revist observations


Discover
Ground
Classify
Evaluate
Measure
Prior Qual AND/OR Usupervised Methods
Search terms
Extend search terms by exploring similar terms
Doc set
Intensive and Extensive reading in and across contexts
Category
Classify using active and distant supervised learning
Classified doc set
Evaluate and Bias correct using test set
Validated measure
Measure category across unites of interest
Noisy Labels
Update search terms
Sample locations based on estimates

From confirmatory and complementary designs to operations

1. **Confirmatory operations** - confirm pattern pattern found with one method with another
2. **Complementary operations** - use different method to investigate different aspects.

How have I designed my analysis to best answer the research question/support the main claim?

1. What is the role of each analysis?
    - What question is it suppose to address?
    - What role does it play in the sequential design of your study?
    - Is it a main or supportive analysis

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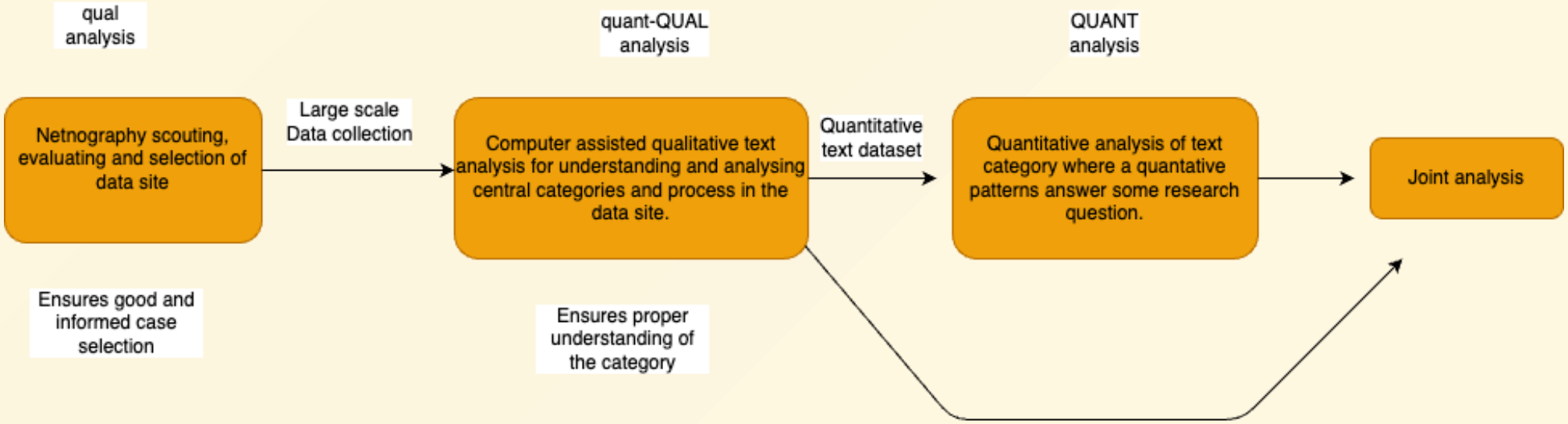
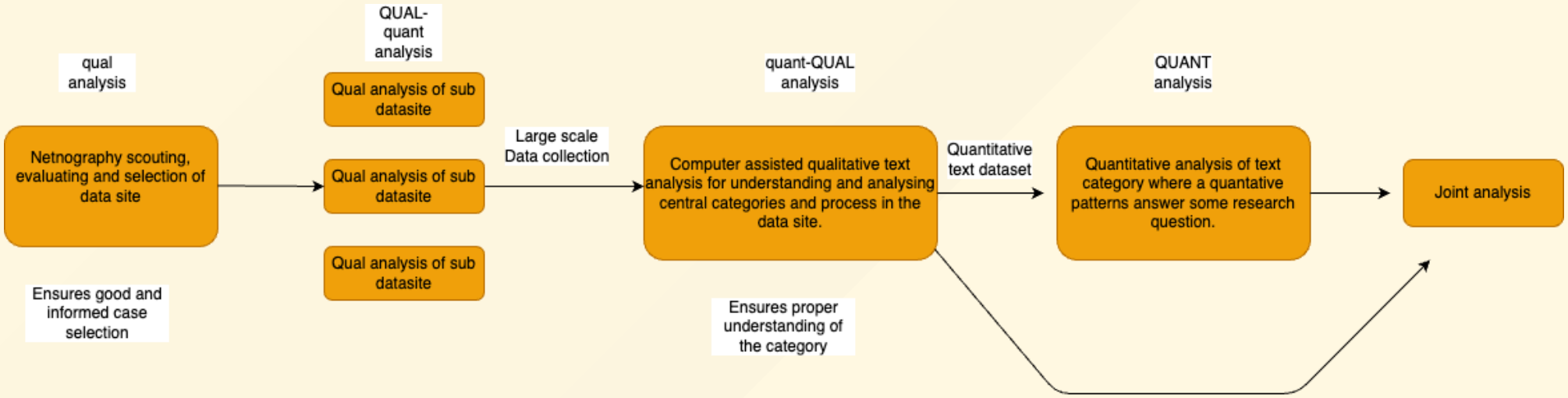
    - What role does it play in the sequential design of your study?
    - Is it a main or supportive analysis
2. What is the overall analytical design and its justification?
    - Is it complementary or confirmatory?
    - How does it help support your main claim?
    - What are the benefits and potential problems given other designs?

Sequencial Analysis Design Simple



Sequencial Analysis Design hierachical



For Tuesday

- Talk about quality of inference in both qual and mixed method.
- Run through the written assignment.
- The presentation for Thursday